
Michael R. Schaar

Machine Learning • Data Science • UX/UI
Design

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EDUCATION

University of California San Diego, La Jolla

*B.S. in Cognitive Science with a Specialization in Machine Learning and
Neural Computation*

Minor in Design

Graduation: June 2025 | Cumulative GPA: 3.658

Relevant Coursework:

- Advanced Machine Learning Methods & Supervised Machine Learning Algorithms
- Brain Computer Interfaces
- Practical Data Science in R
- Data Science with Pandas
- Python
- Systems Neuroscience & Neuroanatomy and Physiology
- Linear Algebra & Vector Calculus
- Cyborgs Now and in the Future
- Sensory Neuroscience & Cognitive Psychology

Key Design Courses:

- Startup Company Studio
- Advanced Interaction Design
- Prototyping and Interaction Design
- Human-Computer Interaction

Palomar College, San Marcos

Computer Science

August 2017 - May 2020 | Cumulative GPA: 3.68

Relevant Coursework:

- Programming Fundamentals with C

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- Differential & Integral Calculus
 - Statistics & Probability

TECHNICAL SKILLS

- **Machine Learning:** Supervised/Unsupervised Learning, Classification, Clustering, Cross-Validation, Model Evaluation, Conditional Random Fields (CRF)
- **Data Science:** Data Wrangling, Data Cleaning, Data Analysis, Data Visualization, Statistical Analysis, Exploratory Data Analysis (EDA)
- **UX/UI Design:** User Research, User Testing, Prototyping, Human-Computer Interaction
- **Programming:** Python, R, React, JavaScript, HTML, CSS, C
- **Libraries:** Pandas, NumPy, scikit-learn, Matplotlib, Seaborn, ggplot2, dplyr, tidyr
- **Tools:** Jupyter Notebook, Figma, Git/GitHub, Visual Studio Code

EXPERIENCE

Jamba Juice, Temecula CA

Team member | September 2017 - March 2019

- Delivered excellent customer service, improving team efficiency during high-traffic hours.
- Managed multitasking in a fast-paced environment, ensuring quality and customer satisfaction.
- Maintained cleanliness, food safety, and operational standards.

PROJECTS

From Pixels to Predictions: A Comparative Study of CRF and Auto-context CRF Models Under Varying Sliding Window Contexts

- Implemented and compared baseline CRF and Auto-context CRF models for Optical Character Recognition (OCR) using varying context window sizes and training dataset splits.
- Analyzed how contextual information influences character recognition accuracy and compared model performance across multiple experimental conditions.

ReWorded: A BCI Project

- Collected and analyzed EEG data to investigate whether N400-related neural responses could differentiate between real words and non-words during language processing tasks.
- Evaluated machine learning models, including SVM, Gradient Boosting, and MLP classifiers, using EEG-derived features to assess real-time word recognition performance.

Tidyclust by Team Tibble

- Implemented k-means clustering with tidyclust in R to explore species-based patterns in the Palmer Penguins dataset.
- Analyzed cluster assignments and centroid characteristics to better understand variation in female penguin bill measurements across species.

Comparative Evaluation of Classifiers on UCI Datasets: A Cross-Validation Approach

- Implemented and compared Logistic Regression, SVM, and Random Forest classifiers on the Iris, Breast Cancer Wisconsin, and Wine Quality datasets using cross-validation.
- Assessed classification accuracy, optimized model hyperparameters, and examined feature importance to evaluate algorithm performance across different data characteristics.

Analyzing the Relationship Between Mean Income & EV Population

- Analyzed electric and hybrid vehicle registration data alongside household income data to investigate factors associated with EV adoption in New York and Texas.
- Applied statistical analysis and data visualization techniques to evaluate relationships between income levels and vehicle registration patterns across these regions.

Pivotour

- Conducted user research, competitive analysis, and service design to develop a startup concept that transforms airport layovers into personalized local travel experiences.
- Created branding, business models, interactive prototypes, and A/B-tested landing pages to evaluate user engagement and support product strategy decisions.

Portfolio.AI

- Researched challenges faced by UX designers when creating professional portfolios and designed an AI-powered platform to streamline the portfolio development process.
- Developed concepts for AI-assisted moodboards, typography recommendations, content organization, and originality analysis to support creativity while preserving individual design identity.

EZ Gym Interactive Kiosk

- Designed and prototyped an interactive fitness kiosk and mobile application concept to help users create, manage, and track personalized workout routines through guided exercise selection and real-time progress monitoring.
- Evaluated usability through user testing and developed features including instructional exercise tutorials, performance analytics, gym navigation, and sensor-assisted repetition tracking.

Assisting Blind People to Orient and Commute

- Applied human-centered design principles to explore how assistive technology can support independent navigation and commuting for blind individuals.
- Designed a conceptual cochlear implant system that combines GPS-based voice guidance with object detection to improve accessibility, situational awareness, and user safety.

Sensory-Friendly Audio and Display Adaptability for Video Playback

- Conducted user research and literature review to evaluate how sensory-friendly audio and display adaptations can improve video playback experiences for individuals with sensory sensitivities.
- Analyzed user feedback on content warnings and adaptive audio/visual settings to identify design considerations related to personalization, accessibility, user control, and digital well-being.

ACHIEVEMENTS

- UC San Diego Provost Honors (2022)
- Palomar College Dean's List (2018)

EXTRACURRICULAR

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- UC San Diego Surf Team Member (2022-2025)